

1. What is a Module? Explain the Need of Modules.

Ans:

A module is a Python file (.py) that contains functions, classes, variables, and statements which can be reused in other Python programs.

Modules help organize code into separate files and make programming easier and more efficient.

Need of Modules

Code Reusability – Write code once and use it multiple times.

Better Organization – Large programs can be divided into smaller files.

Easy Maintenance – Debugging and updating code becomes easier.

Avoid Code Duplication – Reduces repeated code.

Provides Built-in Functions – Python offers modules like `math`, `random`, and `os`.

Example

```
import math
```

```
print(math.sqrt(25))
```

Output:

```
5.0
```

2. State the Different Ways of Importing a Module

Ans:

a) Import Entire Module

```
import math
```

```
print(math.sqrt(16))
```

b) Import Specific Function from a Module

```
from math import sqrt
```

```
print(sqrt(16))
```

c) Import Multiple Functions

```
from math import sqrt, factorial
```

```
print(sqrt(25))
```

```
print(factorial(5))
```

d) Import All Functions

```
from math import *
```

```
print(sqrt(36))
```

e) Import Module with Alias

```
import math as m
```

```
print(m.sqrt(49))
```

f) Import Specific Function with Alias

```
from math import sqrt as s
```

```
print(s(64))
```